

Problem 77 (Silk Road Mathematics Competition 2021 Problem 2). Let n be a positive integer. Let $\{x\} = x - \lfloor x \rfloor$ be the fractional part of a positive real number x . Prove that

$$\left| \{\sqrt{n}\} - \frac{1}{2} \right| > \frac{1}{8(\sqrt{n} + 1)}.$$

Solution. Let $k = \lfloor \sqrt{n} \rfloor$. Then

$$\left| \{\sqrt{n}\} - \frac{1}{2} \right| = \frac{1}{2} |2k + 1 - 2\sqrt{n}| = \frac{1}{2} \cdot \frac{|4k^2 + 4k + 1 - 4n|}{2k + 1 + 2\sqrt{n}}.$$

The numerator of the last fraction is an odd positive integer, and therefore it is at least 1. Thus,

$$\left| \{\sqrt{n}\} - \frac{1}{2} \right| \geq \frac{1}{2} \cdot \frac{1}{2k + 1 + 2\sqrt{n}} \geq \frac{1}{2(4\sqrt{n} + 1)} > \frac{1}{8(\sqrt{n} + 1)}.$$

□

Problem 78 (Japan MO Final 2021 Problem 1). Let $\mathbb{Z}^+$ be the set of all positive integers. Find all function(s) $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ such that $m \mid n$ if and only if $f(m) \mid f(n) - m$ for any positive integers m and n .

Solution. Answer: $f(n) = n$ for any $n \in \mathbb{Z}^+$. Indeed, $m \mid n$ if and only if $m \mid n - m$ trivially holds.

Putting $m = n$ in the given condition, since $n \mid n$, we have $f(n) \mid f(n) - n$. This implies $f(n) \mid n$.

Now we prove that $f(k) = k$ by induction on k . For $k = 1$, this follows from $f(1) \mid 1$. Consider the case $k > 1$ and assume all smaller cases hold. If $f(k) < k$, then $f(f(k)) = f(k)$ by the inductive hypothesis. Together with $f(k) \mid k$, we find that $f(k) \mid f(f(k)) - k$. By the given condition (with $m = k$ and $n = f(k)$), we obtain $k \mid f(k)$. This contradicts $f(k) < k$.

On the other hand, we must have $f(k) \leq k$ since $f(k) \mid k$. Thus, the only possibility is $f(k) = k$. This completes the inductive step, and so $f(n) = n$ is the only solution. □

Problem 79 (Japan MO Final 2021 Problem 2). Let $n > 1$ be an integer. Two players X and Y play the following game on an $n \times 2021$ chessboard. Firstly, X colours every cell black or white. Secondly, Y places a chess piece in one of the cells in the top row, and designates one of the cells in the bottom row as the terminal cell. Then X moves the chess piece in the following way:

- If the chess piece lies in a white cell, X has to move it to the cell below.
- If the chess piece lies in a black cell, X can move it to the left cell or the right cell (if the cell exists), and then move it to the cell below.

Find the smallest possible value of n such that X can always move the chess piece to the terminal cell.

Solution. The answer is $n = 2022$.

First, we prove that $n \geq 2022$. We denote the cell in the i th row and the j th column as (i, j) . In each step, the chess piece can only move from (i, j) to one of $(i + 1, j - 1)$, $(i + 1, j)$ or $(i + 1, j + 1)$. The column index can only be increased by at most 1. Therefore, if Y places the chess piece at $(1, 1)$ and designates the cell $(n, 2021)$ as the terminal cell, then it is impossible for X to win if $2021 - 1 > n - 1$, i.e. $n < 2021$. So it suffices to consider $n \geq 2021$.

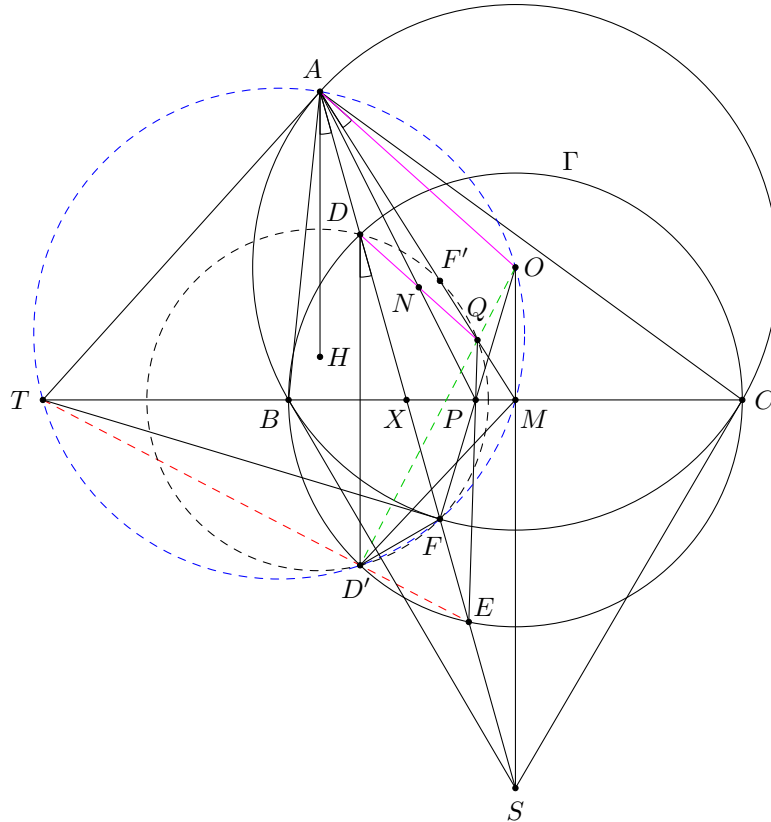
For $n = 2021$, if Y follows the same strategy, then in order that the chess piece can move from $(1, 1)$ to $(2021, 2021)$, it must move along the diagonal line that contains all cells of the form (k, k) . Therefore, all cells (k, k) with $1 \leq k \leq 2020$ must be black. In that case, we show that the chess piece cannot move from $(1, 1)$ to $(2021, 2020)$. Indeed, since all cells (k, k) with $1 \leq k \leq 2020$ are black, the chess piece must move diagonally from $(1, 1)$ to (k, k) for some $2 \leq k \leq 2020$, and then move to $(k + 1, k - 1)$ (since otherwise the chess piece ends up in the cell $(2021, 2021)$). But since $2020 - (k - 1) > 2021 - (k + 1)$, it is impossible for the chess piece to move further to $(2021, 2020)$. This proves our claim, and so we must have $n \geq 2022$.

On the other hand, we show that X has a strategy when $n = 2022$. Indeed, X colours the following cells black (and the other cells white):

$$\begin{aligned}
& (1, 1), (1, 3), (1, 4), (1, 6), \dots, (1, 1009), (1, 1011), \\
& (1, 1013), \dots, (1, 2016), (1, 2018), (1, 2019), (1, 2021), \\
& (2, 2), (2, 5), \dots, (2, 1010), \\
& (2, 1012), \dots, (2, 2017), (2, 2020), \\
& (3, 1), (3, 4), \dots, (3, 1009), \\
& (3, 1013), \dots, (3, 2018), (3, 2021), \\
& (i, j) \text{ for } 4 \leq i \leq 2022, 1 \leq j \leq 2021.
\end{aligned}$$

Due to symmetry, we may assume the initial cell is $(1, a)$ where $1 \leq a \leq 1011$. In the first step, the chess piece can move to one of $(2, 3k + 2)$ where $0 \leq k \leq 336$. Then the chess piece can move to either $(3, 3k + 1)$ or $(3, 3k + 3)$, and then move to either $(4, 3k + 2)$ or $(4, 3k + 3)$. We now show that the chess piece can move diagonally to any cell $(2022, j)$ in the bottom row. The difference in column index is at most $2021 - 2 = 2019$ while the difference in row index is 2018. As X can select $(4, 3k + 2)$ or $(4, 3k + 3)$, we may assume the difference in column index has the same parity as the difference in row index. Thus, the difference in column index cannot exceed the difference in row index, which means the chess piece can move to the cell $(2022, j)$. So we are done. $\square$

Problem 80. Let $\triangle ABC$ be an acute triangle with circumcentre O . The A -symmedian of $\triangle ABC$ meets the circle with diameter BC at the points D and E (where D lies closer to A than E), and meets the circumcircle of $\triangle ABC$ again at F . Let P be the intersection point of BC and FO , and let Q be the intersection of the line EP and the median AM of $\triangle ABC$. Prove that AP bisects the segment DQ .



Solution. We define the following points.

- X is the intersection point of AF and BC
- S is the intersection point of the tangents at B and C to (ABC) (note that S lies on AD as a property of the symmedian)
- T is the intersection point of the tangents at A and F to (ABC) (note that T lies on BC again because $ABFC$ is harmonic)

- N is the intersection point of AP and DQ
- D' is the reflection of D in the line BC
- F' is the reflection of F in the line BC
- H is the orthocentre of $\triangle ABC$

Let $\Gamma = (BECD)$. Using the harmonic quadrilateral $ABFC$, we obtain

$$-1 = (A, F; B, C) \stackrel{A}{=} (T, X; B, C) \stackrel{D}{=} (TD \cap \Gamma, E; B, C).$$

As BC is a diameter of Γ , this implies $TD \cap \Gamma$ and E are symmetric about the line BC . Therefore, T, D', E are collinear.

Next, we prove M, O, A, T, D', F are concyclic. The points M, O, A, T, F are concyclic as $\angle OMT = \angle OAT = \angle OFT = 90^\circ$. For the point D' , since M is the centre of Γ , we have

$$\angle MD'E = 90^\circ - \frac{1}{2}\angle D'ME = 90^\circ - \angle D'DE = 90^\circ - \angle HAE = 90^\circ - \angle OAM = \angle MAT.$$

(Note that $\angle HAE = \angle OAM$ since AO and AM are isogonals of AH and AE with respect to $\angle BAC$ respectively.) This proves D' also lies on (MAT) as desired.

Now, applying Pascal's theorem to $AMTD'OF$, we see that the points $AM \cap D'O$, $MT \cap OF = P$ and $TD' \cap FA = E$ are collinear. It follows that $AM, D'O, PE$ are concurrent at Q . In other words, D', Q, O are collinear.

The next step is to prove that Q lies on $(DD'FF')$ (note that $DD'FF'$ is an isosceles trapezoid). Indeed, we have

$$\angle F'QD' = \angle AOQ + \angle QAO = \angle AOD' + \angle HAE = \angle AFD' + \angle D'DE = 180^\circ - \angle FD'D = 180^\circ - \angle D'DF'.$$

This proves Q, D', D, F' are concyclic. Thus, we obtain

$$\angle DQD' = \angle DFD' = \angle AOD',$$

which implies $AO \parallel DQ$. Lastly, we see that

$$(D, Q; N, \infty_{DQ}) \stackrel{A}{=} (X, M; P, AO \cap BC) \stackrel{O}{=} (X, S; F, A) \stackrel{B}{=} (C, B; F, A) = -1.$$

This shows N is the midpoint of DQ . □